Pricing Baku Real Estate from Scratch: Decision Trees and Pegasos SVMs on bina.az Listings

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`github.com/mmahammad57/binaaz-ml-project`

Problem and dataset

- bina.az, a property portal for Azerbaijan: Kaggle collection of 100,775 sale listings, 51 columns
- 91,314 listings after cleaning, 171 features
- **Task A:** regress log(price); skewness $6.00 \rightarrow 0.28$
- **Task B:** premium if price $\geq$ 215,000 AZN, else standard
- Threshold = median *training* price; validation and test reuse it: 50.3 % premium in every split

Data cleaning and leakage

Presenter: Mahammad Mammadli

- Rows removed: 9,412 missing price, area or rooms; 4 duplicates; 45 outliers outside 10k–5M AZN or 10–2,000 m^2
- 30 columns dropped: `unit_price` and `total_price` give the price away; IDs, URLs, times, sellers and free text identify listings
- Floor "k / m" split into floor, storeys and `has_floor`; land area parsed from "sot"
- 5 flag columns encoded 1 = present, 0 = missing
- Seeded, stratified 70/15/15 split: 63,919 / 13,697 / 13,698, one `get_splits()` for every stage

Decision tree from scratch

Presenter: Ibrahim Aghazade

- Gini or entropy (Task B), variance (Task A); split gain

$$\Delta = I(S) - \frac{|S_L|}{|S|} I(S_L) - \frac{|S_R|}{|S|} I(S_R)$$

- Search: every feature, every *observed* value as threshold; largest Δ wins
- Stop: pure node, max. depth, min. rows to split, min. rows per leaf, min. gain: fixed at 10 / 10 / 5 / 0
- Matches scikit-learn: Task B 464 vs 462 leaves, 99.9 % of test labels agree; Task A predictions correlate 0.9989

Pegasos SVM from scratch

Presenter: Sahib Aliyev

- Objective, labels ± 1:

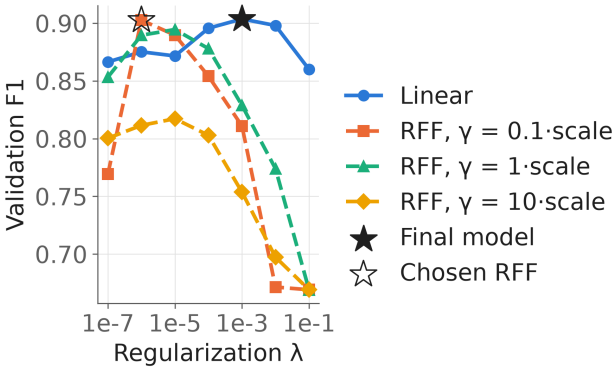
$$J(\mathbf{w}, b) = \frac{\lambda}{2}\|\mathbf{w}\|^2 + \frac{1}{n}\sum_i \max\bigl(0,\, 1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b)\bigr)$$

- One random example per step, $\eta_t = 1/(\lambda t)$, then project $\mathbf{w}$ onto $\|\mathbf{w}\| \le 1/\sqrt{\lambda}$:

$$\mathbf{w} \leftarrow (1 - \eta_t \lambda)\,\mathbf{w} + \eta_t\, \mathbf{1}\bigl[y_i(\mathbf{w}^\top \mathbf{x}_i + b) < 1\bigr]\, y_i \mathbf{x}_i$$

- Margin $2/\|\mathbf{w}\|$; support vectors: rows with $y_i(\mathbf{w}^\top \mathbf{x}_i + b) \le 1$
- RFF, $D = 500$: 256 MB instead of a 32.7 GB kernel matrix

Tuning the SVMs on validation



7 λ from 10^{-7} to 10^{-1}; scale $= 1/171$

- Rule: within 0.005 of the best F1, largest λ
- Final: linear, $\lambda = 10^{-3}$, F1 0.904, CV 0.897 ± 0.003
- RFF, $\lambda = 10^{-6}$: F1 0.903, CV 0.763 ± 0.095
- It needs 10^6 updates; a CV fold gives 320,000

Benchmark against scikit-learn (test set)

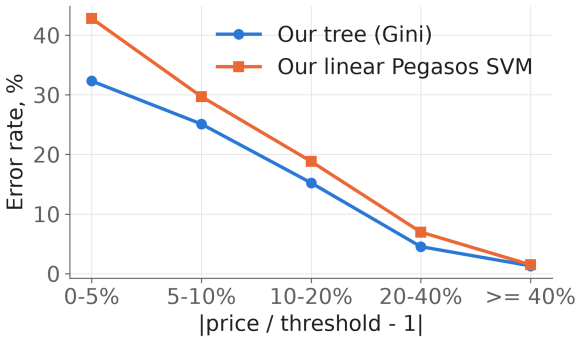
Presenter: Khizri Sakhrayev

- Task A, R^2 on log price: ours 0.853, scikit-learn 0.852

Task B	F1	F1, CV	Fit (s)
Tree Gini (ours)	0.926	0.907 ± 0.006	86.8
Tree Gini (sklearn)	0.926	0.907 ± 0.006	0.31
Pegasos linear (ours)	0.904	0.897 ± 0.003	2.7
SGD hinge (sklearn)	0.905	0.894 ± 0.004	0.29

- Our tree: $277\times$ slower; fit time grows $\times 15.0$ (ours) vs $\times 10.4$ (sklearn) from 2k to 16k rows

Where the models fail



Task B error by distance from the threshold

- Within 5 % of the 215,000 AZN threshold: tree 32.3 %, SVM 42.8 % wrong
- That band holds 28 % of tree errors
- Task A median error: 14.3 % cheapest, 19.6 % dearest decile
- Districts with 1–49 vs 1,000+ training rows: 20.0 vs 11.4 %

Limitations and honest notes

- Trees not tuned: fixed 10 / 10 / 5, while the SVMs were tuned
- RFF Pegasos is unstable on less data: CV F1 0.763 ± 0.095
- CV on 20,000 of 77,616 rows, because our tree is $277\times$ slower
- One-hot built on all rows: 2 columns all-zero in training
- 0 means "not applicable": floor for 15,338 listings, land area for 76,126

Contributions and next steps

Mahammad Mammadli	data pipeline: cleaning, leakage, split
Ibrahim Aghazade	decision tree and its scikit-learn test
Sahib Aliyev	Pegasos SVM, RFF, sweep and plot tools
Khizri Sakhrayev	metrics, benchmark, tuning, analysis, report

- Faster split search: sorted values with running sums
- Then a real tree sweep; bagging or boosting of our trees
- Location features: distance to the centre, district statistics